

Solution Architecture — Script-Controlled ACL (Institution Details)

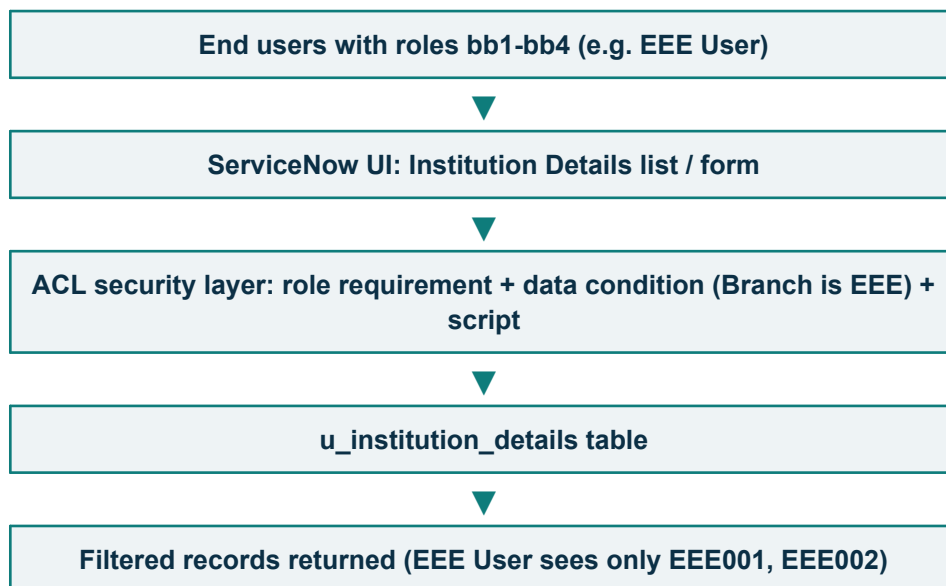
Phase 3 · Project Design Phase

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This document describes the solution architecture for the Script-Controlled ACL project on ServiceNow PDI dev443713. It shows how a request from a branch end-user flows through the ServiceNow UI into the Access Control List security layer, where a role requirement, a data condition and a server-side script together decide access to rows in the `u_institution_details` table, and how filtered records are returned. All layers are native ServiceNow platform components; there is no external application, framework or database.

Platform	ServiceNow Now Platform (Australia / Zurich family release)
Instance	dev443713.service-now.com (Personal Developer Instance)
Security engine	Access Control List (ACL) with server-side GlideScript (gs.hasRole)
Core table	u_institution_details with Branch choice field (ECE / EEE / CSE)

Architecture Flow



Component Responsibilities

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End users and roles (bb1 read, bb2 create, bb3 write, bb4 delete)	Identify the requester and the operation attempted. The test user EEE User holds all four bb roles; admin (System Administrator) always passes; Abel Tuter holds none.
ServiceNow UI / Institution Details list	Presents the list and form for u_institution_details and issues the read, create, write or delete request; renders only what the ACL layer permits, or the "Security constraints prevent access" message when denied.
ACL security layer (sys_security_acl)	For each operation, evaluates the required role, then the data condition, then the script. Read is Advanced with the condition Branch is EEE (u_branch=EEE); Create, Write and Delete check only their bb role.
READ ACL script (gs.hasRole)	Server-side GlideScript: returns true if the user has role admin, true if the user has role bb1, and false otherwise, enforcing the field-value-plus-role decision for reads.
u_institution_details table	Stores the Institution Details records (Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description) including ECE001, EEE001, EEE002 and CSE001; returns only rows that satisfy the ACL evaluation.
Supporting configuration (Impersonation, security_admin, default role removal)	security_admin elevation enables editing ACLs; Impersonation verifies each persona's view; removing the auto-generated u_institution_details_user role ensures every ACL depends only on its bb role.